

The Assistant Erased You: Measuring Loss of Authorship Signals in AI-Mediated Communication

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ABSTRACT

Research on AI-mediated communication has examined how AI assistance shapes interpersonal perceptions and reduces stylistic diversity across users. We ask a complementary question at the individual level: after a message is rewritten by an AI writing assistant, can its author still be distinguished from others? We introduce the *Idiolect Erasure Rate* (IER), defined as the reduction in authorship-attribution accuracy following AI-assisted rewriting. We evaluate IER on three pre-generative-AI corpora using a stylometric model and the authorship-specific LUAR model. Heavy rewriting substantially weakens authorship signals in personal blogs and workplace email, reducing LUAR attribution by as much as 66.5 percentage points, but has a much smaller effect on topic-structured news, where topic remains predictive of authorship. Additional analyses suggest that rewriting produces stylistic convergence despite substantial semantic overlap, and that content-sensitive attributers understate the loss captured by authorship-specific models. Heavily rewritten messages may also evade AI-text detectors, making them difficult both to attribute to their human authors and to identify as AI-assisted, a phenomenon we call *double erasure*. IER measures computational attributability rather than human recognition, and we release it as an open and reproducible protocol for evaluating authorship-signal loss in AI-mediated communication.

1 INTRODUCTION

AI writing assistants increasingly mediate everyday written communication by rewriting messages for clarity, professionalism, or tone before they are sent. Research on AI-mediated communication (AIMC) has examined how such assistance affects trust [10, 12], responsibility [5], users’ beliefs [13], and stylistic diversity across populations [1, 17]. We ask a complementary question at the level of the individual: after a message passes through an AI writing assistant, can its author still be distinguished from others?

Prior work in adversarial stylometry has shown that writers can deliberately modify their language to evade computational attribution [2, 4, 16]. We study a different setting. Rather than intentionally concealing identity, AI writing assistants may weaken authorship signals as an unintended consequence of rewriting text for clarity, fluency, or tone.

Writing carries recurring stylistic patterns, including word choice, sentence rhythm, and punctuation, that together form an individual’s *idiolect*. These patterns enable computational authorship-attribution systems to distinguish one writer from another [15]. AI rewriting may alter these signals while preserving much of the underlying message, raising the question of whether a person’s writing remains computationally attributable after assistance.

To study this phenomenon, we introduce the *Idiolect Erasure Rate* (IER), defined as the reduction in authorship-attribution accuracy following AI-assisted rewriting. We evaluate IER across three communication registers using both stylometric and neural attribution models. Our results show that AI rewriting substantially weakens recoverable authorship signals in interpersonal writing, while having much smaller effects where topical information remains highly predictive of authorship.

Although we measure computational attribution rather than human recognition, we argue that changes in attributable writing style represent an overlooked dimension of AI-mediated communication. Understanding when AI assistance preserves or suppresses these signals has implications for future identity-preserving writing assistants and for how AI mediation is evaluated more broadly.

Our contributions are threefold:

- (1) We introduce the *Idiolect Erasure Rate* (IER), an open protocol for quantifying the loss of computational authorship signals after AI-assisted rewriting.
- (2) We show that AI writing assistants substantially weaken authorship signals in interpersonal communication while leaving topic-driven attribution largely intact in topic-structured writing.
- (3) We identify *double erasure*: AI-assisted text may simultaneously become difficult to attribute to its human author and difficult to identify using current AI-text detectors [16, 19].

2 RELATED WORK

2.1 AI-mediated communication

AI-mediated communication (AIMC) studies how AI systems modify, augment, or generate messages on a person’s behalf [8]. Prior work has shown that AI mediation influences interpersonal trust [10, 12], responsibility attribution [5, 9], collaboration [10], and the opinions users express [13]. More recently, researchers have examined how people perceive AI-assisted writing and its implications for authenticity [11, 14]. Our work examines a complementary question: whether AI assistance changes the computationally attributable signals that distinguish one writer from another.

2.2 Authorship attribution and idiolect

Computational authorship attribution exploits recurring stylistic patterns, including function-word preferences, sentence rhythm, and punctuation, to identify writers across anonymous texts [15, 18]. A related literature studies authorship obfuscation, in which writers deliberately modify their style to evade attribution [2, 4]. Our setting differs in both intent and mechanism.

Rather than intentionally hiding identity, we study whether everyday AI-assisted rewriting unintentionally weakens computationally measurable authorship signals. We therefore adapt attribution accuracy as a way of quantifying stylistic erosion rather than recovering authorship.

2.3 From stylistic homogenization to individual identity

Recent work has shown that AI writing assistants reduce stylistic diversity across users [1, 3, 17], while recursive training similarly reduces diversity within language models [21]. These studies characterize changes at the population level.

Communication research likewise argues that identity is expressed through language and negotiated with an audience [6, 7, 11]. We complement these perspectives by examining whether AI rewriting weakens the computational signals through which an individual writer remains distinguishable. Rather than measuring homogenization across a population, we quantify the loss of attributable writing style for each author.

3 METHODOLOGY

3.1 The Idiolect Erasure Rate

The *Idiolect Erasure Rate* (IER) quantifies how much AI-assisted rewriting weakens computational authorship signals. We first train an authorship-attribution model on each author’s original writing, then evaluate whether it can still identify the same authors after their messages have been rewritten by an AI assistant. A larger reduction in attribution accuracy indicates that the assistant has weakened the stylistic cues on which the attributer relies.

Formally, for attributer f , assistant g_c under rewriting condition c , and held-out human messages $\{x_i\}$ with corresponding authors $\{a_i\}$,

$$\text{IER}(g_c, f) = \text{acd}(f(\{x_i\}), \{a_i\}) - \text{acd}(f(g_c(x_i)), \{a_i\}),$$

where IER is the percentage-point reduction in attribution accuracy after AI-assisted rewriting.

We report results using both a **surface** (stylometric) attributer and a **deep** (neural) attributer, treating agreement and disagreement between them as informative. Because attribution performance depends on both the evaluation setting and the attribution method, IER is not an intrinsic property of an AI assistant but of the assistant, rewriting condition, attributer, and corpus considered together.

3.2 Experimental setup

Corpora. We evaluate IER on three corpora representing different communication registers. The *Blog Authorship Corpus* [20] contains informal, personal, topic-diverse writing (50 authors, 200 held-out messages), making it representative of the register targeted by AI writing assistants and one in which topic is only weakly associated with author identity. The *Enron Email Corpus* consists of authentic interpersonal email (20 users, 80 held-out messages), representing a more formulaic communication register. The *Reuters C50* corpus contains news articles in which each journalist writes within a fixed topical beat (25 authors, 100

held-out messages), providing a control condition where topic strongly correlates with authorship. All three corpora predate modern generative AI systems.

Rather than truncating documents, we retain messages at their natural length (mean 190 words, capped at 400). Truncation artificially reduces attribution performance and produces incomplete fragments that AI assistants partially reconstruct, introducing an additional confound.

Attributers. We evaluate both a *surface* and a *deep* authorship attributer. The surface model uses TF-IDF character (2–4)-grams and word (1–2)-grams with a linear SVM. The deep model is LUAR [18], which represents each author by the mean embedding of their training documents and assigns authorship using nearest-profile retrieval.

A style-sensitive attributer is essential for measuring idiolect erosion. MiniLM is largely insensitive to word-order perturbations (accuracy drops only from 0.385 to 0.360 after random word shuffling), indicating that it relies primarily on semantic content. In contrast, LUAR drops from 0.710 to 0.205 under the same manipulation, demonstrating substantially greater sensitivity to stylistic information. We therefore use LUAR as our primary deep attributer and report MiniLM only as a topic-sensitive baseline.

Assistant. Our primary rewriting model is the local open-weight model Qwen2.5-1.5B-Instruct, decoded greedily to ensure deterministic and reproducible outputs. Our evaluation pipeline is model-agnostic and also supports commercial AI assistants (§4).

Rewriting conditions. We evaluate three prompting conditions: *light*, which corrects grammar and spelling only; *heavy*, which rewrites text for clarity and professionalism; and *preserve*, which improves writing while explicitly preserving the author’s voice. All prompts are released with the code.

Evaluation protocol. For each author, a disjoint set of messages is held out for evaluation, while the remaining documents are used for training, providing the attributer with as much data as each corpus permits (blogs: mean 119 posts per author; LUAR profiles average 60 training documents). IER compares attribution accuracy on the held-out original messages with accuracy on their AI-rewritten counterparts.

Authorship attribution is evaluated in the closed-set setting, where every test message belongs to one of the known authors. This provides an upper bound on attribution performance, whereas real-world recognition is generally open-set. Chance performance is therefore the reciprocal of the number of authors.

Authors are selected according to document availability. Random author subsampling yields comparable attribution baselines (0.80–0.88 on the blog corpus), indicating that this selection strategy does not materially inflate performance.

Statistical significance is assessed using McNemar’s exact test. We report 95% bootstrap confidence intervals and, for headline results, cluster-bootstrap confidence intervals over authors. To facilitate reproducibility, we release all prompts, random seeds, LUAR checkpoints, and author splits.

4 RESULTS

Strong authorship signals before AI assistance. Before rewriting, both surface and deep attributers recover authors far above chance across all three corpora (Table 1). Surface attribution

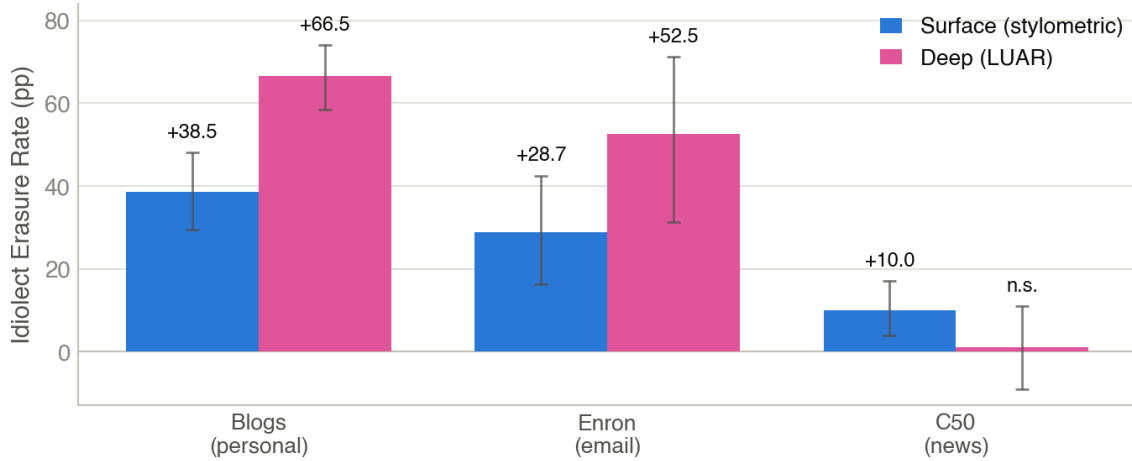


Figure 1: Heavy-rewrite IER across three corpora (matching Table 1), for the surface (stylometric) and deep (LUAR) attributers, with author-clustered 95% CIs. Surface attribution is significantly erased on every register; the deep fingerprint is erased strongly on both interpersonal corpora (Blogs +66.5, Enron +52.5) but is null on beat-structured news (C50), where topic stands in for identity. Bars are significant ($p < 0.001$) unless marked n.s.

Table 1: Heavy-rewrite IER in percentage points, with the baseline attribution accuracy it is measured from (base). Surface is stylometric; deep is LUAR. Post-rewrite accuracy is base – IER/100. *: $p < 0.001$; n.s.: not significant. $n = 200$ (blogs), 80 (Enron), 100 (C50).

Corpus	chance	surface		deep (LUAR)	
		base	IER	base	IER
Blogs (personal)	.02	.810	+38.5*	.815	+66.5*
Enron (email)	.05	.938	+28.7*	.713	+52.5*
C50 (news)	.04	.910	+10.0*	.710	+1.0 (n.s.)

reaches 0.810, 0.938, and 0.910 on Blogs, Enron, and Reuters C50, respectively, while LUAR achieves 0.815, 0.713, and 0.710. In the closed-set setting, aggregating multiple original messages identifies the correct author with high reliability (Fig. 2). The central question is how much of this authorship signal survives AI-assisted rewriting.

Surface authorship signals weaken consistently. Heavy AI rewriting significantly reduces surface attribution across all three corpora (Fig. 1; Table 1). The largest reduction occurs on the Blog corpus (IER = +38.5 points), followed by Enron (+28.7) and Reuters C50 (+10.0). Grammar-only rewriting produces substantially smaller effects, suggesting that extensive stylistic rewriting, rather than simple error correction, is primarily responsible for erasing surface authorship cues.

Deep authorship signals are substantially more vulnerable. The effect is even stronger for deep attribution. Under heavy rewriting, LUAR loses 66.5 percentage points on Blogs and 52.5 on Enron, corresponding to the loss of more than three quarters of the recoverable authorship signal. The effect grows monotonically with rewriting intensity, and even prompts that explicitly

instruct the assistant to preserve the author’s voice remove most of the recoverable signal. These findings remain significant under author-level cluster bootstrapping.

The observed loss reflects stylistic convergence. Several analyses indicate that the measured erasure reflects changes in writing style rather than changes in semantic content. First, rewritten texts become substantially less distinguishable from one another, indicating convergence toward more similar stylistic representations. Second, semantic similarity between original and rewritten messages remains high despite large reductions in attribution accuracy. Third, function-word attribution, which minimizes lexical-content information, also exhibits substantial degradation. Together, these results suggest that AI rewriting primarily weakens stylistic rather than semantic cues.

The choice of attributer matters. IER depends strongly on the attribution model. Although the stylometric model and LUAR achieve nearly identical baseline accuracy, LUAR exhibits substantially larger erasure after rewriting, indicating that IER depends on the type of authorship signal an attributer captures rather than its overall accuracy. In contrast, content-sensitive models such as MiniLM consistently underestimate erasure because they rely more heavily on semantic information than on stylistic patterns.

The Reuters C50 corpus illustrates this distinction. Although LUAR achieves strong baseline attribution performance, heavy rewriting produces little deep erasure because topic remains highly predictive of authorship. Function-word attribution nevertheless declines substantially, indicating that stylistic signals are weakened even when topical information continues to support attribution.

The effect generalizes across AI assistants. Comparable levels of deep authorship erasure are observed for commercial assistants, including GPT-4o-mini and Gemini Flash, as well as across multiple sizes of the Qwen2.5 family. The phenomenon therefore does

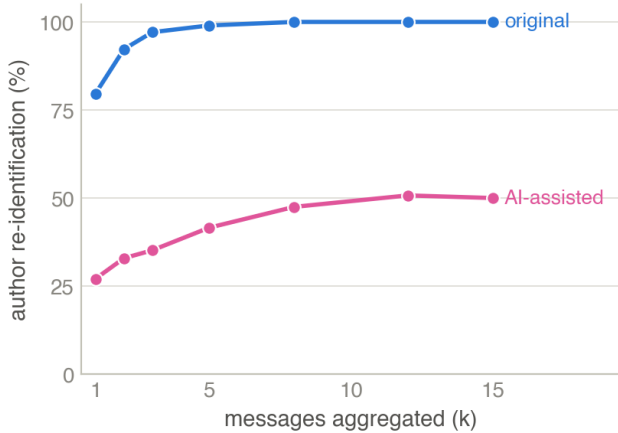


Figure 2: Population accumulation (heavy rewrite, LUAR, 20 blog authors). Re-identification from pooled messages reaches 100% by $k=8$ for originals but saturates near 50% for assisted messages, so aggregation does not recover erased identity.

not appear to depend on a particular model architecture or parameter scale.

Identity loss accumulates across correspondence. Aggregating more rewritten messages does not recover a writer’s identity. Whereas attribution from original messages approaches perfect accuracy after observing several messages, attribution from rewritten messages saturates at approximately 50% and shows little improvement thereafter (Fig. 2). This suggests that idiolect erasure is best understood as an accumulation across a person’s correspondence rather than repeated degradation of an individual message.

Voice-preservation prompts are insufficient. Explicitly instructing the assistant to preserve the author’s voice reduces surface-level erasure but leaves most of the deep authorship signal unrecovered. Simple prompting therefore appears insufficient to preserve computationally identifiable writing style, motivating future work on identity-preserving AI writing assistance.

Robustness checks. All reported effects survive a Bonferroni correction across our tests (only C50 light does not, $p = 0.031$), and every significant IER excludes zero under author-clustered bootstrapping (deep intervals [+58, +74] on Blogs, [+31, +71] on Enron). Grammar-only (light) rewriting erases far less than heavy rewriting (surface +5.0 on Blogs; +3.7, n.s., on Enron), and a non-generative grammar-correction baseline removes only +15 LUAR points versus +66.5 for generative rewriting, so the effect is specific to generative rewriting rather than any edit. Enron’s high baseline is partly signature leakage: stripping sign-off names and quoted text lowers baselines (surface 0.94 to 0.85, LUAR 0.71 to 0.58) but the erasure persists (+30/+39, $p < 10^{-6}$), while the same stripping leaves Blog baselines unchanged. Exact commercial IERs (heavy, LUAR) are GPT-4o-mini +58.5/+53.8 and Gemini +60.5/+52.5 on Blogs/Enron.

5 DISCUSSION

Our findings suggest that computational authorship signals constitute an overlooked dimension of AI-mediated communication, with implications for the design and evaluation of AI writing assistants.

Design implications. IER provides both a design objective and an evaluation metric for identity-preserving AI writing assistance. Rather than relying on prompting alone, future systems could explicitly optimize for preserving computational authorship signals while improving writing quality. Our *preserve* condition suggests that prompting alone is insufficient, particularly for deeper stylistic representations. More generally, IER could serve as a measurable “fingerprint budget” that allows users to control how much of their writing style is retained.

Disclosure alone is not enough. Labels such as “AI-assisted” indicate provenance, not whether a message still reflects its author’s distinctive style. Moreover, heavily rewritten messages may evade current AI-text detectors [16, 19], producing what we term *double erasure*: text that is difficult both to attribute to its human author and to identify as AI-assisted.

Implications for AIMC. Existing AIMC research has examined how AI mediation affects trust, responsibility, beliefs, and stylistic diversity. Our results suggest a complementary perspective: AI assistance may also weaken the attributable stylistic signals that distinguish one writer from another. We therefore propose idiolect preservation as an additional dimension of AI-mediated communication. Its desirability, however, is context-dependent: preserving a recognizable voice may support authenticity and accountability, while reduced attributability may benefit privacy or anonymity. Future work should therefore examine how systems can balance writing quality, user control, and authorship preservation across different communication settings.

6 LIMITATIONS AND CONCLUSION

IER measures computational attributability rather than human recognition. We interpret reductions in attribution as evidence that measurable authorship signals have weakened, not as direct evidence that familiar readers recognize writers less accurately. Future work should therefore combine computational evaluation with human-recognition studies and compare AI-assisted rewriting against edit-matched human editing.

Our experiments evaluate three AI assistants across three communication registers and datasets containing 20–50 authors. Although broader evaluations would strengthen external validity, the consistency of the observed effects across interpersonal writing, attribution methods, and rewriting conditions suggests that idiolect erasure is not specific to a particular model or dataset.

Taken together, our findings show that AI-assisted rewriting can substantially weaken recoverable authorship signals in interpersonal communication while promoting stylistic convergence across writers. We introduce the Idiolect Erasure Rate (IER) as a reproducible framework for quantifying this phenomenon and hope it supports future research on designing AI writing assistants that better preserve individual writing style.

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